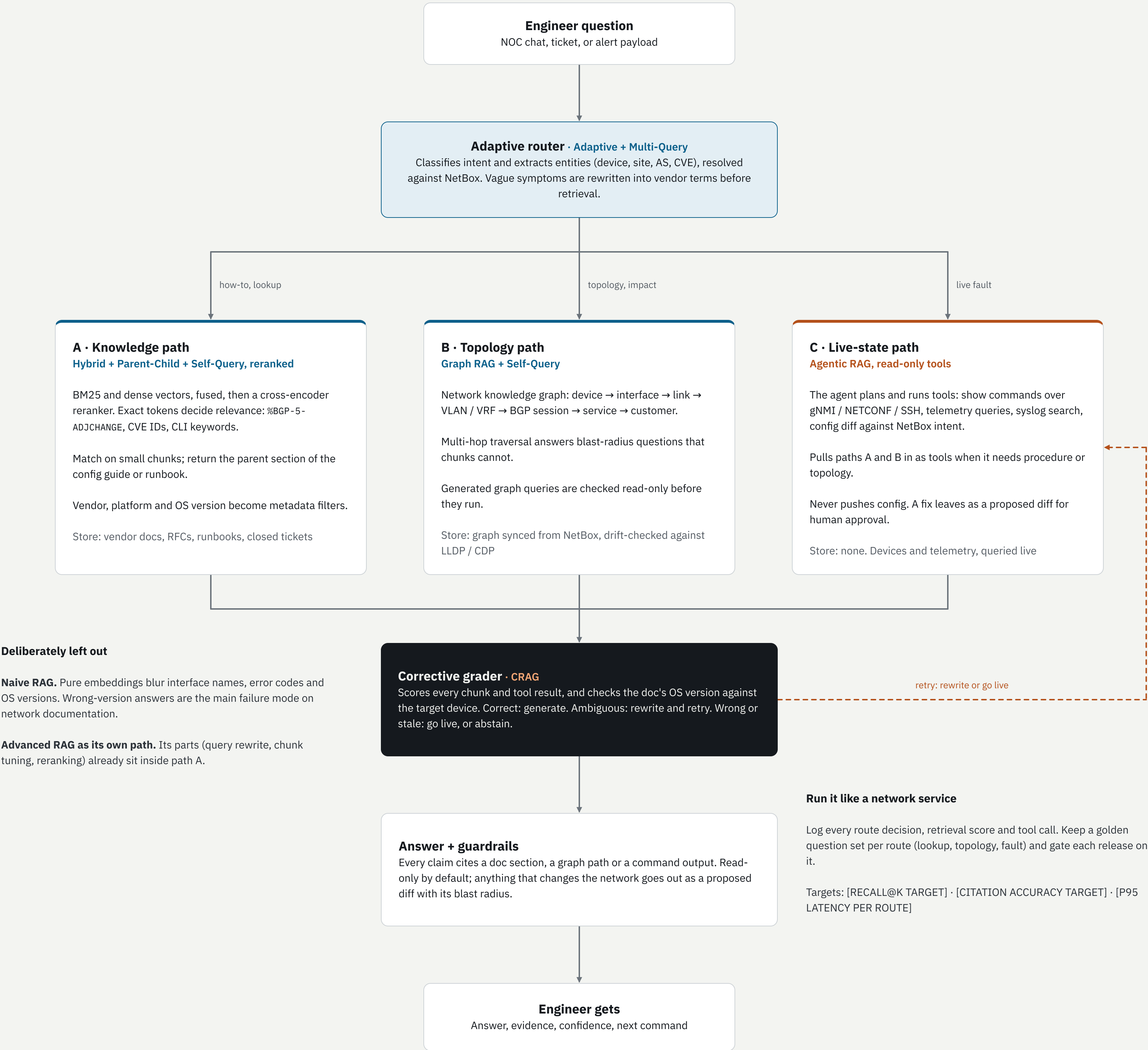


One router, three retrieval paths, one grader

Route each question to the retrieval that fits its data: hybrid search for documentation, a knowledge graph for topology, and a read-only tool-using agent for live state. A corrective grader checks every answer before an engineer sees it.



Eight of the ten are used, one is folded in, one is dropped

The ten are not alternatives on one list. They sit on four axes: how you index, how you retrieve, how you transform the query, and how you control the loop. Pick per axis, driven by a property of network data.

TYPE	AXIS	STATUS	WHERE	WHY, FOR NETWORKING
Hybrid RAG	Retrieval	Core	Path A	BM25 catches show ip bgp summary, %LINK-3-UPDOWN, CVE IDs. Dense vectors catch "the neighbor keeps flapping". Neither alone covers both.
Graph RAG	Index	Core	Path B	"What breaks if core-02 goes down?" is a multi-hop walk over links, sessions and services. No chunk holds that answer.
Parent-Child RAG	Index	Core	Path A	Config guides are long and nested. Match one step precisely, then hand the model the whole procedure with its prerequisites.
Self-Query RAG	Query transform	Core	Paths A, B	Questions name a device; NetBox supplies its vendor, platform, OS version, site and role. Apply those as metadata filters before similarity, not after.
Multi-Query RAG	Query transform	Core	Router	Engineers describe symptoms; vendors document mechanisms. "Flapping" becomes "adjacency change", "hold timer expired", "BFD session down".
Adaptive RAG	Control flow	Core	Router	Most questions are lookups. Only faults need live tools. Routing keeps the common case fast and cheap.
Corrective RAG	Control flow	Core	Grader	A guide for the wrong OS train reads as relevant and is wrong. Grade against the target device, then retry, go live, or abstain.
Agentic RAG	Control flow	Core, gated	Path C	Root cause needs live state the index will never have. Tools are read-only; any write needs a human.
Advanced RAG	Pipeline tuning	Folded in	Path A	Its parts (rewrite, chunk tuning, reranking) are used inside path A. It is a label for those parts, not a separate path.
Naive RAG	Baseline	Dropped	—	Keep it only as the evaluation baseline. Embeddings alone miss exact tokens and ignore OS version.

"Why is BGP to AS 65010 down on edge-rtr-01?"

A live fault uses all three paths. The agent gathers state, the graph supplies context, the docs supply the procedure, and the grader drops what does not match the device.

1	Route	Intent: live fault. Entities: device <code>edge-rtr-01</code> , peer AS 65010. Sent to path C, with A and B available as tools.	Adaptive
2	Resolve context	Graph lookup (synced from NetBox): vendor, platform and OS version of the device; the peer's interface, circuit ID and provider; services and customers riding that circuit.	Graph + Self-Query
3	Collect live state	Read-only calls: BGP neighbor detail, interface counters and errors, BFD session state, syslog for the peer over the last hour.	Agentic
4	Rewrite the symptom	Evidence says "hold timer expired" with the interface up. Rewritten into vendor terms: BGP hold timer expiry, BFD down, MTU mismatch on the peer link.	Multi-Query
5	Retrieve procedure	Hybrid search filtered to the device's vendor and OS train. Child chunk matches one check; the full troubleshooting section is returned. Past tickets on the same circuit come with it.	Hybrid + Parent-Child
6	Grade	The guide for the device's OS train is kept. A ticket from an older train with different BFD defaults is dropped. If nothing passes, the answer says so instead of guessing.	CRAG
7	Answer	Likely cause, each claim tied to a command output or doc section, the next command to run, and the customers affected from the graph. A fix, if any, arrives as a proposed diff for approval.	Guardrails